



International University of Sarajevo

Faculty of Engineering and Natural Sciences

Embedded Systems

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REPORT

Parking Sensor System

Professor: Tarik Namas

Students: Hena Šehović

Berina Juković

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1. Introduction

1.1 History of Parking Sensor Systems

Parking sensor systems, or proximity sensors or reverse assist systems, have seen a revolution in the past two decades. Initially used to enhance the safety and ease of parking, these appeared on the scene in the late 1970s and early 1980s. The initial systems were restricted to high-end luxury vehicles and employed simple ultrasonic or infrared technology.

The first commercial parking sensors were often ultrasonic-type systems mounted on the rear bumper of vehicles. (Picture 1) The systems would emit sound waves and measure the time for the echo to return, which helped drivers sense obstructions at their backs. Toyota was among the first car makers to introduce this technology, as it placed an ultrasonic sensor system in its 1983 Crown model.



Figure 1: Parking sensor on a fender

With advancing technology in the 1990s and the 2000s, parking sensors have become more mainstream and affordable, eventually becoming a feature in the majority of mid-range and compact cars. Modern systems now use a combination of ultrasonic sensors, radar, and cameras to provide more accurate and complete awareness of surroundings. In addition, integration with vehicle infotainment systems and displays has streamlined user experience and provided more information.

Today, parking sensors are not limited to rear sensing but are even popular at the front and sides of vehicles, adding features like automatic parking and collision avoidance. Their development reflects the overall trend towards more automation and driver-assistance functions in modern cars.

1.2 About the System and Current Technologies

Parking sensor systems are a common feature in modern cars, employed to help drivers not bump into any object during low-speed operations like parking and reversing. The systems provide responses through auditory signals, visual signals, or display outputs, improving safety as well as convenience.

Most parking systems employ ultrasonic sensors, emitting high-frequency sound waves and measuring the time for the echo to return once it reflects off an object. It allows the system to detect distance and warn the driver that they are getting too close. The remaining technologies include radar sensors for longer range detections, infrared sensors for nighttime, and camera systems providing a real-time image with extra features like obstacles being tracked and lines of trajectory. The more sophisticated systems make use of sensor fusion, which combines multiple technologies together for greater precision and automatization.

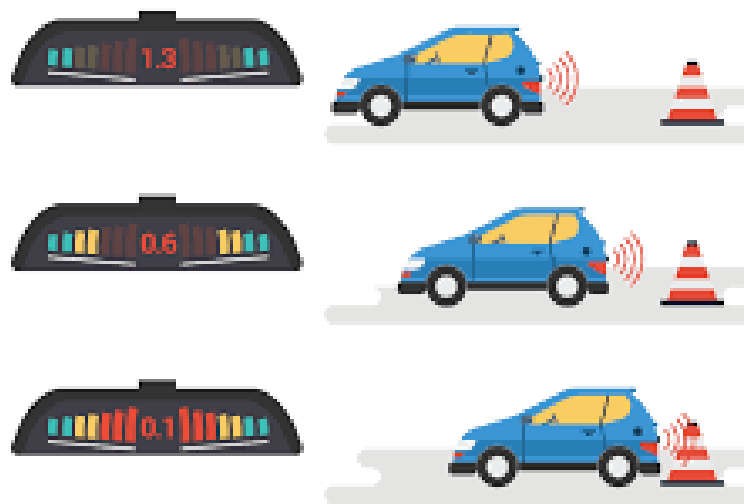


Figure 2: System employing ultrasonic sensors

This project utilizes a simple ultrasonic sensor interfaced with an STM32 Nucleo microcontroller to detect distance and warn the user. A buzzer that rises in frequency as objects approach, a red LED as a visual alert, and an OLED screen to display the precise distance make up the warning system. Although straightforward, this arrangement mimics the fundamental operation of commercial systems and provides a versatile foundation for adding enhancements such as multi-sensor arrays or automatic actions in the future.

1.3 Overview of the Project

The goal of this project was to design and deploy a simple parking sensor system using the STM32 Nucleo microcontroller. The system ought to be capable of assisting drivers by detecting near-by obstacles and providing real-time feedback through sound, light, and display. It mimics the basic functionality of modern automotive parking assistance systems.

At the center of the system is an ultrasonic sensor that continuously measures the distance between the vehicle and any obstacle behind it. The ultrasonic sensor sends ultrasonic waves and calculates the distance from the time taken by the returned wave to reach back. This data is calculated by the STM32 Nucleo board, which acts as the central controller.

Depending on the distance measured, the system operates in three modes:

- A buzzer emits warning beeps whose frequency increases as the object nears.
- A red LED is lit as a visual warning when an obstacle within a predefined range is detected.
- An OLED display shows the actual distance, offering precise visual feedback.

The project uses basic electronic components and embedded C programming in STM32CubeIDE. It demonstrates some elementary ideas such as sensor interfacing, real-time signal processing, and output control in the embedded system field. The result is a functional but easy-to-understand prototype for a parking assistance system.

1.4 Growing Up of the Idea

The idea for this project arose from everyday annoyance drivers face when attempting to park in cramped or crowded spaces. Since we were developing embedded systems for class, the idea of combining real-world sensing with immediate feedback came across as functional and intriguing. We sought to create something functional that is also representative of the basics we were learning—sensor interfacing, signal processing, and control of hardware.

Parking sensors were a feasible and worthwhile project. They are found on new vehicles, but simple enough to be duplicated at the most fundamental level. The selection of using an ultrasonic sensor was because it was inexpensive, simple, and reliable for short-range use. We wished to integrate real-time distance sensing with simple output modes like sound (buzzer), light (LED), and display (OLED) so that the system will be easy to comprehend and react to.

With every passing minute, as we progressed on the idea further, we chose the STM32 Nucleo board for its robust features, ease of adaptability, and compatibility with educational environments. It allowed us to implement the system with ease using STM32CubeIDE and access GPIOs needed to drive various components. The project kept changing from a preliminary idea to an operational prototype that illustrates real-world usage as well as basic embedded system principles.

2. The Project

2.1 Components used

This section describes all the key components required in the design and implementation of the parking sensor system. Each component plays a vital role in ensuring the system works effectively, from sensing the environment to providing audio-visual feedback to the user. The combination of the STM32 microcontroller, sensors, lights, and hardware in enablement makes an efficient and reliable parking aid solution.

1. STM32 Nucleo Microcontroller Board

The STM32 Nucleo board is the core of the project and is based on the ARM Cortex-M microcontroller family. It provides a flexible development platform with multiple GPIO pins, timers, and communication interfaces that are essential for interfacing sensors and output devices. The board is programmed using STM32CubeIDE and supports real-time control and processing, making it ideal for embedded applications such as parking sensors.



Figure 3: STM32 Nucleo Microcontroller Board

2. Breadboard

A breadboard is used to build and test the circuit without soldering. It allows for easy connections and modifications during prototyping. Components and jumper wires are plugged into the breadboard to create a temporary but stable electrical circuit.

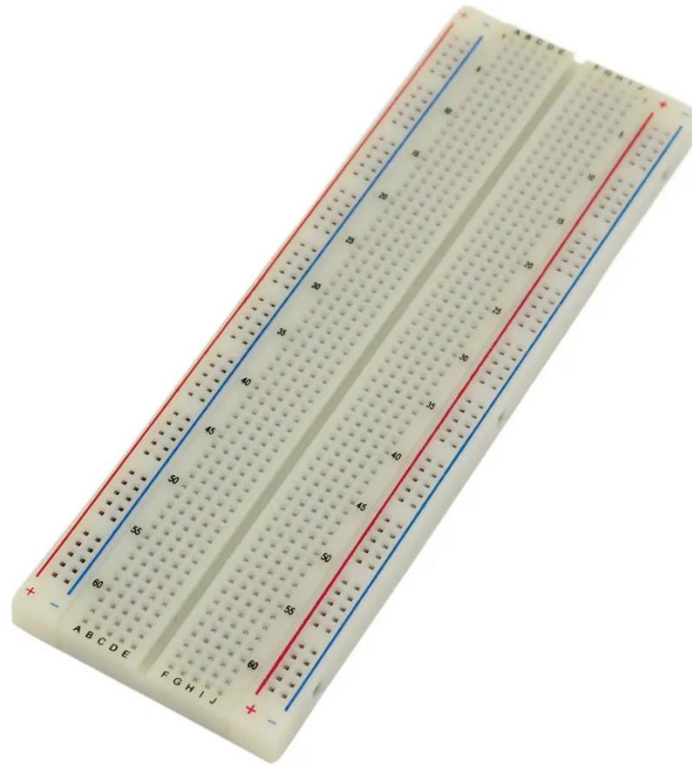


Figure 4: Breadboard

3. Jumper Wires (Male-to-Male, Male-to-Female, Female-to-Female)

Jumper wires of various types are used to connect the microcontroller pins to sensors, actuators, and other components on the breadboard. Male-to-male wires connect pins on the breadboard, male-to-female connect microcontroller pins to the breadboard, and female-to-female wires connect sensors or modules with header pins.



Figure 5: Male-to-Male, Male-to-Female, Female-to-Female Jumper Wires

4. LED

The LED acts as a visual alert indicator. It lights up when an obstacle is detected within a certain distance, providing immediate feedback to the driver. The LED is connected with a current-limiting resistor to ensure safe operation.



Figure 6: LED

5. Buzzer

The buzzer provides an audio alert to the user. In this project, it emits beeping sounds that increase in frequency as the vehicle approaches an obstacle. This audible feedback helps the driver gauge the proximity of nearby objects.



Figure 7: Buzzer

6. Resistors

Resistors are used to limit the current flowing through certain components, such as the LED or sensor inputs, preventing damage due to excessive current. They help maintain safe and stable operation of the circuit.

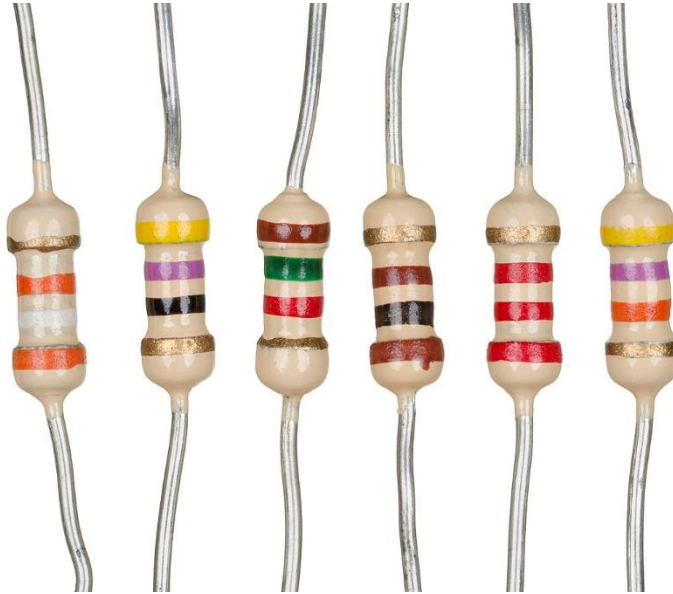


Figure 8: Resistors

7. OLED Display

The OLED (Organic Light Emitting Diode) display provides a visual output that shows the distance measured by the ultrasonic sensor. It is a small, low-power display capable of rendering clear numerical data, allowing the driver to see precise distance readings in real time.



Figure 9: OLED Display

8. Ultrasonic Sensor

The ultrasonic sensor measures the distance between the vehicle and obstacles by emitting ultrasonic waves and calculating the time it takes for the echo to return. This sensor provides accurate distance measurements, which are critical for triggering the buzzer, LED, and updating the OLED display.



Figure 10: Ultrasonic Sensor

2.2 System Architecture / Design Overview

The parking sensor system is designed to detect obstacles with the help of an ultrasonic sensor and provide instant feedback to the user in terms of audio, visual, and display outputs. The system architecture consists of three main functional blocks: sensing, processing, and output.

Sensing Block:

The ultrasonic sensor emits high-frequency sound waves and measures how long it takes the sound waves to return in echoes after they hit objects nearby. The time measured is converted to distance, which is the distance an object is from the vehicle.

Processing Block:

The output block is made up of the STM32 Nucleo microcontroller as a central processing unit. It receives echo signals from the ultrasonic sensor, calculates the distance, and processes this information using embedded firmware. The timers and GPIO ports of the microcontroller are configured to count pulse widths accurately and control output devices.

Output Block:

Based on the measured distance, the system activates a red LED and buzzer to alert the user. The buzzer beeps at a frequency that increases as the vehicle gets closer to an obstacle, providing intuitive audible alerts. Simultaneously, the OLED display shows the exact distance reading in numeric form, providing the user with discrete visual feedback.

2.3 Hardware Implementation

The hardware configuration of the parking sensor system is focused on the master controller, which is the STM32 Nucleo development board. (Picture 3) All the components are mounted on this board using a breadboard with jumper wires of male-to-male and male-to-female to allow for flexibility and ease of prototyping.

The ultrasonic sensor (HC-SR04) is connected to the STM32 via two GPIO pins: one for triggering the ultrasonic pulse and another for receiving the echo. A red LED and a buzzer are also connected to GPIO pins, each with a series current-limiting resistor to avoid burning out the components. The buzzer beeps at increased frequency as the obstacle distance is decreased, and the red LED provides a visual warning when an object is within the critical distance.

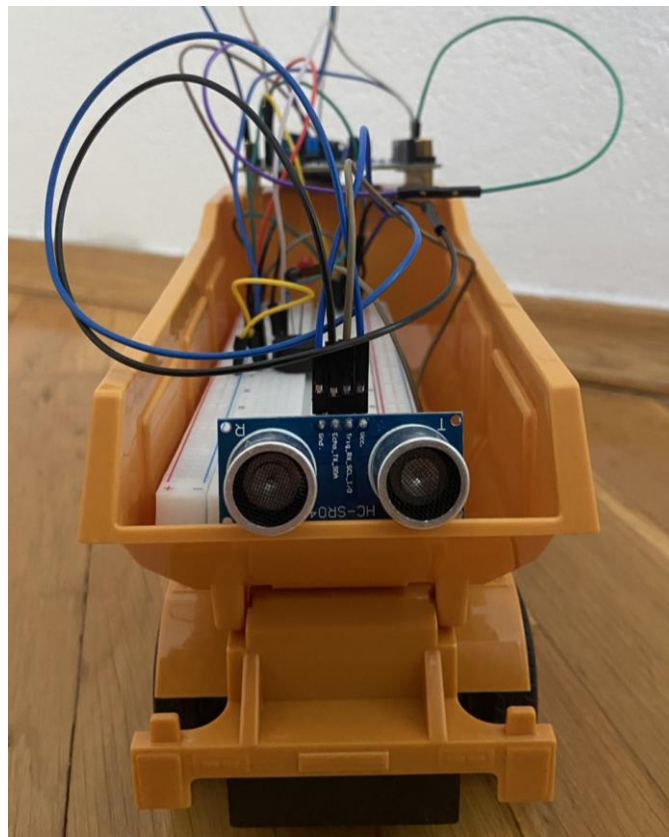


Figure 11: Ultrasonic sensor connected

For the display of live distance readings, an OLED screen (I2C protocol) is employed. It is interfaced to the STM32 board through the I2C pins (SCL and SDA), enabling effective two-wire communication. The system is powered by the USB interface to the STM32 board, which provides 3.3V or 5V depending on the requirement of each device.

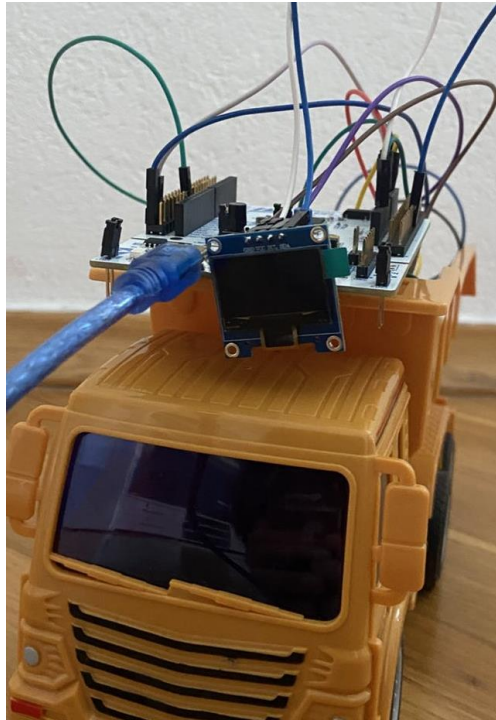


Figure 12: OLED Display connected

Each of the connections was tested separately before implementing the system. The ease of utilization through a breadboard made debugging and changing very easy during development. This setup well demonstrates how sensor input, signal processing, and various types of output can be integrated into an embedded system minimized.

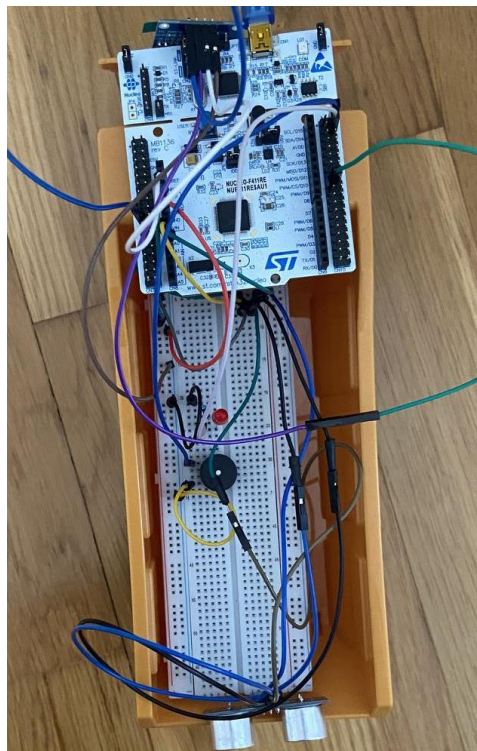


Figure 13: Hardware implementation of the project

2.4 Software Implementation

Implementation of parking sensor system software was done using STM32CubeIDE, which is an integrated development environment tool used for configuring peripherals, embedded C coding, and debugging on the STM32 microcontroller. The software design is modular for maintaining it and making future changes easier.

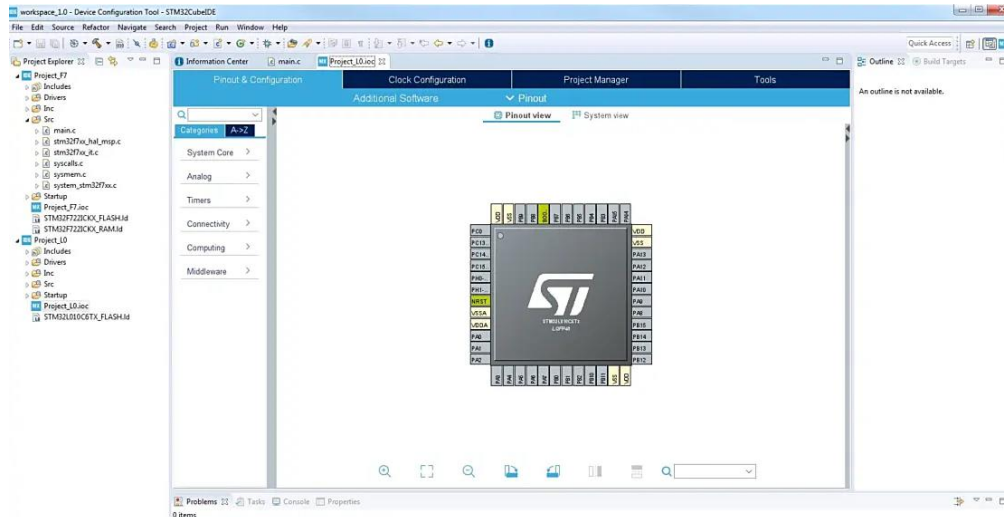


Figure 14: STM32CubeIDE

The parking sensor system uses several General-Purpose Input/Output (GPIO) pins on the STM32 microcontroller, each configured for specific functions. The initialization process ensures that these pins operate correctly for interfacing with external components such as the ultrasonic sensor, buzzer, LED, and OLED display. (see Picture 15)

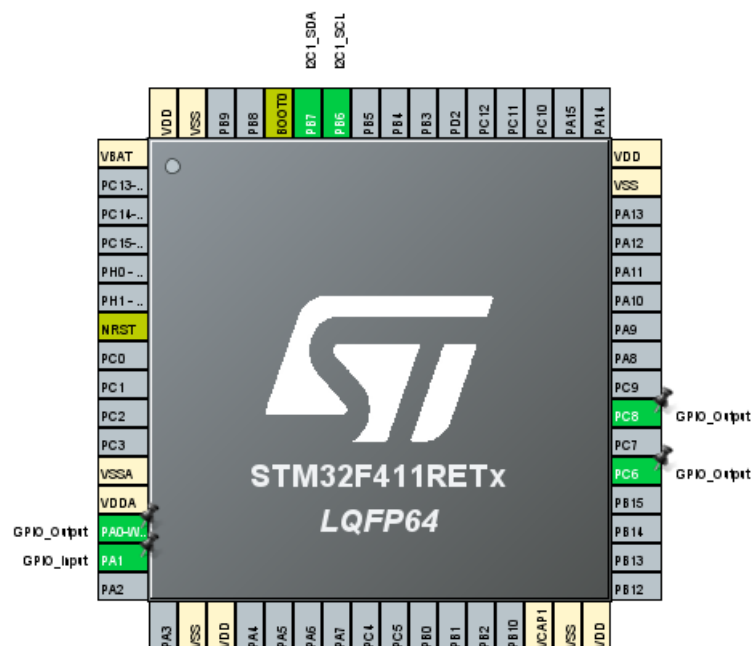


Figure 15: Initializing the pins

The primary function is distance measurement using an ultrasonic sensor. The microcontroller generates a consistent 10µs trigger pulse to initiate the ultrasonic burst. The system waits for the echo pin for the return signal. A timer captures the difference in time between the rising and falling edges of the echo pulse, which is the time-of-flight of the ultrasound. It then converts this pulse duration to distance using the formula: $\text{Distance} = (\text{Speed of Sound} \times \text{Time}) / 2$. It incorporates a timeout to handle the case where no echo is found, either because of an out-of-range situation or sensor failure.

```
while (1)
{
    // 1. Trigger ultrasonic pulse (PA0)
    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, GPIO_PIN_RESET);
    HAL_Delay(2);
    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, GPIO_PIN_SET);
    HAL_Delay(1); // ~10 us
    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, GPIO_PIN_RESET);

    // 2. Wait for echo start (PA1 goes HIGH)
    uint32_t startTick = HAL_GetTick();
    while (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_1) == GPIO_PIN_RESET) {
        if (HAL_GetTick() - startTick > 50) break; // timeout
    }
}
```

Figure 16: Triggering ultrasonic sensor with pulse and echo start

The system provides multi-modal feedback based on the measured distance. The OLED display shows the current distance measurement in real-time. An audible buzzer produces beeps with increasing frequency as the object gets closer to the sensor, providing intuitive distance feedback. A red LED activates when the measured distance falls below a critical threshold, serving as a visual warning. These feedback mechanisms work together to create an effective user interface for parking assistance.

The main control loop executes measurements at regular intervals (approximately 10 times per second) to provide real-time feedback while avoiding sensor interference from previous pulses. The software uses HAL delay functions for basic timing and the microcontroller's timer peripherals for precise measurement of the echo pulse width. This combination ensures accurate distance measurements while maintaining responsive system behavior.

```

// Distance-based blink/beep on PC2 and PC3
if (distance < 10) {
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_SET);
    HAL_Delay(100);
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_RESET);
    HAL_Delay(100);
} else if (distance < 20) {
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_SET);
    HAL_Delay(300);
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_RESET);
    HAL_Delay(300);
} else if (distance < 40) {
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_SET);
    HAL_Delay(600);
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_RESET);
    HAL_Delay(600);
} else {
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_2 | GPIO_PIN_3, GPIO_PIN_RESET);
    HAL_Delay(100);
}

```

Figure 17: What to do depending on the distance ultrasonic sensor gives

2.5 Testing and Results

Upon integrating the hardware and software, the system was mounted on a small truck model to test it in a more practical application. (see Picture 12) All the components, including the STM32 Nucleo board, ultrasonic sensor, buzzer, red LED, and OLED display, were securely mounted on the truck and linked through a breadboard. Power was supplied through Power bank , and the truck was controlled remotely to simulate reversing motion.

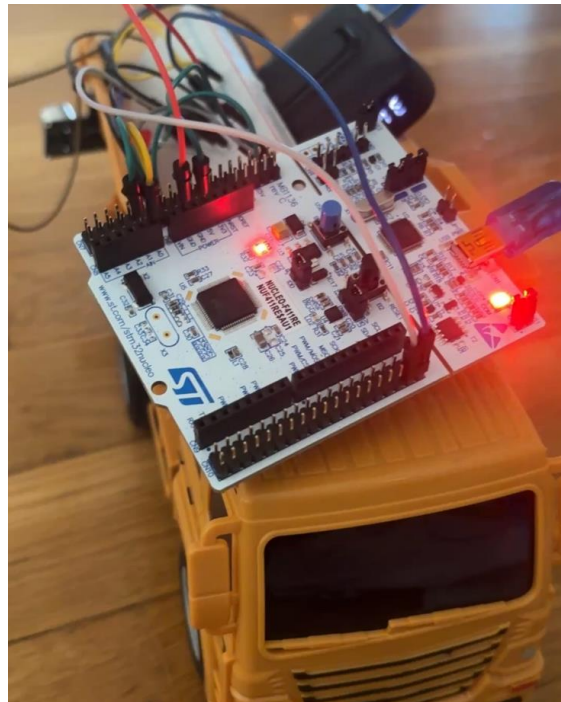


Figure 18: Testing model

During testing, the truck was driven into various objects such as walls and boxes to check the response of the system. The ultrasonic sensor accurately read distances, and the OLED display provided real-time feedback in centimeters. The buzzer sounded at a varying frequency based on the proximity of the truck to the object, while the red LED illuminated when the object came within a predefined danger zone (e.g., less than 30 cm).

This moving test setup demonstrated that the system works well under dynamic conditions. It responded instantly to distance changes, and all the components performed as expected over a number of trials. The remote-controlled truck setup gave us practical insight into how such systems work in real cars, confirming the efficiency and accuracy of our implementation.

2.5 Challenges and Solutions

There were several challenges faced at the design stage of the parking sensor system, both hardware and software. The major challenge was measuring the correct distance using the ultrasonic sensor. The measurements initially were not accurate due to noise and poor timing. This problem was overcome by accurately calibrating the delay functions, using timers to have more precise pulse measurement, and adding software filters to smooth out variations.

The other issue was with buzzer and LED responsiveness. Delay in the code of earlier versions caused responses to occur very slowly over different distances. We fixed this by restructuring the code according to non-blocking logic and timer-based updates, such that the system would respond in real time without any interruption.

We also faced I2C communication problems when integrating the OLED display. The screen would hang or display wrong data at times. This was fixed by checking pull-up resistors on SDA and SCL lines, ensuring a stable voltage level, and adding timeout/error handling subroutines in the code.

An engineering problem was encountered when mounting the system onto the truck. Providing stable connections while in motion necessitated hardening of wires and stabilizing the breadboard. Testing the system on uneven terrain enabled us to identify loose contacts and correct them.

These problems not only improved the system but also intensified our understanding of debugging procedures and embedded system stability under real-world conditions.

3. Conclusion and Future Work

This project successfully exhibited a functional parking sensor system using the STM32 Nucleo microcontroller. Equipped with an ultrasonic sensor for distance measurement, a buzzer for audible feedback, a red LED for visual warning, and an OLED display for displaying the distance, we were successful in mimicking the basic functionality of commercial parking assist systems. The system performed reliably in stationary and moving tests, especially when mounted on a remote-controlled truck, which simulated real-world usage scenarios.

The project reinforced key embedded systems concepts such as peripheral configuration, sensor interfacing, real-time processing, and output control. It also provided valuable practical experience in system integration and debugging.

For the future development, the system can be expanded by the addition of multiple ultrasonic sensors for side and front detection, wireless communication through Wi-Fi or Bluetooth for transmitting data to a mobile app, or the addition of camera modules and object classification using machine learning. The casing and power supply system can also be improved for better mobility and outdoor testing for longer durations.

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Connecting and using the OLED Explanation
<https://www.youtube.com/watch?v=z1Px6emHIeg>

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